

May 1, 2026

Conservative vector fields and domain

reminder: A vector field $\vec{F}: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is called conservative if there is a potential function $f: X \rightarrow \mathbb{R}$ such that $\nabla f|_{\vec{a}} = \vec{F}(\vec{a})$ for all $\vec{a} \in X$.

Theorem: Suppose $\vec{F}: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a C^1 vector field, where as usual we write $\vec{F}(x_1, \dots, x_n) = (F_1(x_1, \dots, x_n), F_2, \dots, F_n)$.

Then:

1) cff $\vec{F}$ is conservative, then $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all i, j in $1, \dots, n$.

1*) cff for at least one pair i, j in $1, \dots, n$

we have $\frac{\partial F_i}{\partial x_j} \neq \frac{\partial F_j}{\partial x_i}$, then $\vec{F}$ is not conservative.

2) cff all the mixed partial derivatives are equal

(that is, $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$ for all i, j in $1, \dots, n$)

X is convex, then $\vec{F}$ is conservative.

and

e.g. Inverse square law

$$\vec{F}: \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}^n$$

$$\vec{F}(\vec{x}) = \frac{c}{\|\vec{x}\|^3} \vec{x} \quad (c \text{ constant})$$

$$f: \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$$

$$f(\vec{x}) = \frac{-c}{\|\vec{x}\|} \quad (\|\vec{x}\| = \sqrt{\sum_{i=1}^n x_i^2})$$

$$\nabla f = (-c(\frac{1}{2}(\sum_{i=1}^n x_i^2)^{-3/2} 2x_1, \dots, -c(\frac{1}{2}(\sum_{i=1}^n x_i^2)^{-3/2} 2x_n)$$

$$\nabla f = \frac{c}{\|\vec{x}\|^3} (x_1, \dots, x_n) = \frac{c}{\|\vec{x}\|^3} \vec{x} = \vec{F}(\vec{x})$$

is conservative (because it admits a potential), but domain is not convex

e.g. Vector line integral of $\vec{F}(x, y) = (\frac{1}{x^2 y}, \frac{1}{x y^2})$ along the line segment (1,1) to (5,8)

$$\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = \frac{-1}{x^3 y^2} - \frac{-1}{x^2 y^3} = 0 \text{ but domain } (\vec{F}) = \{(x, y) : x \neq 0 \text{ and } y \neq 0\} \text{ is not convex}$$

Nevertheless, the line segment C is contained in $X = \{(x, y) : x > 0 \text{ and } y > 0\}$

and X is convex $\implies$ we can find a potential for $\vec{F}: X \rightarrow \mathbb{R}^2$ (maybe not in all the domain)

$$\int \frac{1}{x^2 y} dx = \frac{-1}{x y} + c(y), \quad f(x, y) = \frac{-1}{x y}, \quad \int_C \vec{F} \cdot d\vec{s} = f(5, 8) - f(1, 1) = \frac{-1}{40} - \frac{-1}{1} = 1 - \frac{1}{40}$$

Parametrized surfaces

Recall: To get all the way to discussing Green's Theorem:
 we started with parametrized lines, then parametrized paths,
 orientation preserving or reversing reparametrization,
 then closed simple curves and oriented curves as boundaries of ^{nice} regions in $\mathbb{R}^2$.

We want to do the same moving up by 1 dimension:

Parametrized planes, parametrized surfaces,
 orientation of a surface, reparametrization

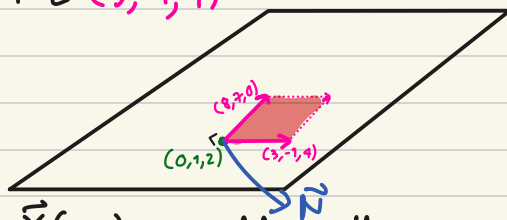
^{nice} oriented surfaces as boundaries of ^{nice} regions in $\mathbb{R}^3$
 and oriented closed simple curves in $\mathbb{R}^3$ as boundaries of ^{nice} surfaces in $\mathbb{R}^3$.

Parametrized plane: $\vec{X}(s,t) = \vec{a} + s\vec{v}_1 + t\vec{v}_2$ $s, t \in \mathbb{R}$
^{point in the plane}
 $\vec{a}, \vec{v}_1, \vec{v}_2$ constant vectors in $\mathbb{R}^3$ with $\vec{N} = \vec{v}_1 \times \vec{v}_2 \neq \vec{0}$
^{vectors parallel to the plane, but not parallel to each other}

e.g. $\vec{X}(s,t) = (0,1,2) + s(8,7,0) + t(3,-1,4)$

$$\vec{T}_s = (8,7,0) \quad \vec{T}_t = (3,-1,4)$$

$$\vec{N} = (8,7,0) \times (3,-1,4)$$



def/notation: For the parametrized plane $\vec{X}(s,t)$ we define the
 vectors $\vec{T}_s = \frac{\partial \vec{X}}{\partial s}$, $\vec{T}_t = \frac{\partial \vec{X}}{\partial t}$ and $\vec{N} = \vec{T}_s \times \vec{T}_t = \frac{\partial \vec{X}}{\partial s} \times \frac{\partial \vec{X}}{\partial t}$
^{two vectors tangent to the plane, but not parallel to each other}
^{normal vector to the parametrized plane}

The parallelogram on the surface formed by the vectors $\vec{T}_s = \frac{\partial \vec{X}}{\partial s}$ and $\vec{T}_t = \frac{\partial \vec{X}}{\partial t}$
 has area $\text{Area}(\square) = \|\vec{T}_s \times \vec{T}_t\| = \|\vec{N}\|$

def: Let D be a region in $\mathbb{R}^2$ consisting of an open connected set (and possibly) together with some or all of its boundary points.

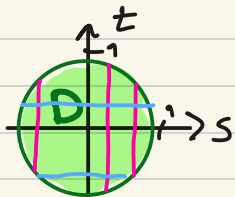
A parametrized surface in $\mathbb{R}^3$ is a continuous function

$\vec{X}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ which is one-to-one on all D , except possibly on ∂D (boundary).

$S = \vec{X}(D) \subseteq \mathbb{R}^3$ is called a surface and $\vec{X}$ its parametrization.

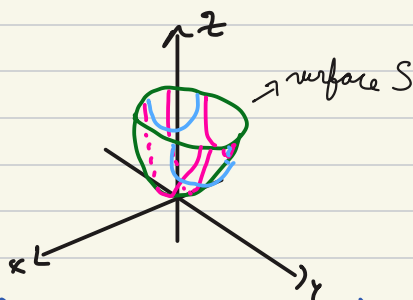
If the parametrization $\vec{X}: D \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is C^1 and the vector-valued function $\vec{N}(s,t) = \frac{\partial \vec{X}}{\partial s} \times \frac{\partial \vec{X}}{\partial t}$ is never $\vec{0}$, then we say the parametrized surface $S = \vec{X}(D)$ is smooth.

e.g. $D = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$



$$\begin{aligned} \vec{X}(s,t) \\ \parallel \\ (s, t, s^2 + t^2) \end{aligned}$$

$$\begin{aligned} \vec{T}_s &= \frac{\partial \vec{X}}{\partial s} = (1, 0, 2s) \\ \vec{T}_t &= \frac{\partial \vec{X}}{\partial t} = (0, 1, 2t) \end{aligned}$$

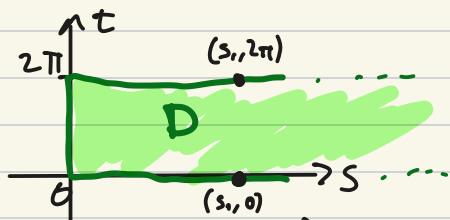


$$\vec{N} = \vec{T}_s \times \vec{T}_t = (-2s, -2t, 1) \neq \vec{0} \text{ for any } s, t$$

This is an example of a C^1 parametrization of a smooth surface

def: For each constant s_0, t_0 , the curves $\vec{X}(s_0, t)$ and $\vec{X}(s, t_0)$ are called coordinate curves.

e.g. $\vec{X}(s, t) = (s \cos t, s \sin t, s^2) \quad s \geq 0$
 $0 \leq t \leq 2\pi$



Let's check if this is
 a diff. param. of a smooth
 surface

We know $\vec{X}$ is C^1 , is $\vec{X}$ one-to-one?

iff $\vec{X}(s_1, t_1) = \vec{X}(s_2, t_2)$, then either $(s_1, t_1) = (s_2, t_2)$
 or (s_1, t_1) and (s_2, t_2) are in ∂D

$$(s_1 \cos(t_1), s_1 \sin t_1, s_1^2) = (s_2 \cos(t_2), s_2 \sin t_2, s_2^2)$$

$$\Rightarrow s_1^2 = s_2^2 \Rightarrow s_1 = s_2$$

$$\leadsto s_1 (\cos t_1, \sin t_1) = s_2 (\cos t_2, \sin t_2)$$

$$t_1 = t_2 + \text{a multiple of } 2\pi$$

$\Rightarrow$ only possibility is that t_1, t_2 are either 0, 2π

Up to here we know this is a parametrized surface

$$\vec{T}_s = (\cos t, \sin t, 2s) \quad \vec{T}_t = (-s \sin t, s \cos t, 0)$$

$$\vec{N} = \vec{T}_s \times \vec{T}_t = (-2s^2 \cos t, -2s^2 \sin t, s \cos^2 t - (-s \sin^2 t))$$

$$= (-2s^2 \cos t, -2s^2 \sin t, s) \text{ is only } \vec{0} \text{ if } s=0 \text{ (in } \partial D)$$

smooth (except at the boundary)